Advanced DevOps

ASSIGNMENT BOOK

(50 Real-World Scenarios)

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Advanced Assignment Book (50 Real-World Scenarios)

Assignment 1 — Multi-Environment Terraform Architecture

Scenario:
Design reusable Terraform code to deploy separate environments (dev, staging, prod).

Requirements:

- Use modules.
- Separate state files per environment.
- Use remote backend (S3 + DynamoDB locking).
- Use workspace or folder-based strategy.
- Implement tagging strategy.
- Implement environment-specific variables.

Assignment 2 — Highly Available Web Tier (Terraform)

Provision:

- 2 EC2 instances in different AZs
- Application Load Balancer
- Auto Scaling Group

- CloudWatch alarms for scaling

Ensure zero downtime rolling updates.

Assignment 3 — Secure VPC Architecture

Design a VPC with:

- Public and Private subnets
- NAT Gateway
- Bastion Host
- Private EC2 instances
- Restrictive security groups

Document traffic flow.

Assignment 4 — Terraform State Recovery

Simulate:

- Accidental deletion of Terraform state.
- Recover infrastructure using import.
- Handle drift detection.

Provide step-by-step documentation.

Assignment 5 — Blue/Green Deployment (AWS + Kubernetes)

Deploy a Node.js app using:

- EKS
- ALB
- Two versions (v1, v2)
- Gradual traffic switch

Rollback on failure.

Assignment 6 — GitHub Actions CI/CD with Multi-Stage Docker

Pipeline must:

- Run tests
- Scan with Trivy
- Build Docker image
- Push to ECR
- Deploy to Kubernetes
- Fail on vulnerability severity HIGH+

Assignment 7 — Self-Healing Kubernetes App

Deploy an app that:

- Crashes randomly
- Must restart automatically

- Configure liveness & readiness probes
- Show logs proving self-healing

Assignment 8 — Horizontal & Vertical Scaling

Implement:

- HPA (CPU-based)
- VPA (if supported)
- Load testing to trigger scaling
- Document scaling behavior

Assignment 9 — Kubernetes RBAC Isolation

Create:

- Namespace per team
- Role allowing read-only access
- RoleBinding per user
- Prevent cross-namespace access

Assignment 10 — Ingress with TLS

Deploy:

- NGINX Ingress Controller

- Let's Encrypt certificate
- Domain routing
- HTTPS redirection

Assignment 11 — Centralized Logging Stack

Deploy:

- EFK (Elasticsearch, Fluentd, Kibana)
- Collect logs from all pods
- Filter logs by namespace
- Create dashboard

Assignment 12 — Prometheus Monitoring Stack

Install:

- Prometheus
- Grafana
- Node Exporter
- Custom application metrics

Create CPU alert rule.

Assignment 13 — Python S3 Sync Tool (Advanced)

Enhance upload tool to:

- Compare local vs S3 checksums
- Resume interrupted uploads
- Handle multi-threading
- Auto-delete removed files

Assignment 14 — Disaster Recovery Simulation

Simulate:

- Node failure
- AZ failure
- etcd corruption

Recover cluster without downtime.

Assignment 15 — Infrastructure Cost Optimization

Analyze AWS usage and:

- Identify idle EC2
- Identify unattached volumes
- Generate cost report
- Propose optimization strategy

Assignment 16 — CI/CD with Manual Approval Gate

Implement pipeline:

- Build
- Security Scan
- Deploy to staging
- Manual approval
- Deploy to production

Assignment 17 — Docker Security Hardening

Harden Dockerfile:

- Non-root user
- Multi-stage builds
- Minimal base image
- Remove unnecessary packages

Assignment 18 — Kubernetes Network Policies

Implement:

- Deny all traffic
- Allow frontend → backend only
- Block backend → frontend

Verify using curl tests.

Assignment 19 — Canary Deployment

Implement:

- 10% traffic to new version
- Monitor error rate
- Promote automatically

Assignment 20 — Vault Integration

Store secrets in HashiCorp Vault.

Inject secrets into Pods securely.

Assignment 21 — Terraform Module Registry

Create reusable module:

- EC2
- Security group
- EBS

Publish internally.

Assignment 22 — Kubernetes Stateful Application

Deploy:

- MySQL StatefulSet
- Persistent storage
- Stable network identity
- Backup mechanism

Assignment 23 — GitOps with ArgoCD

Setup:

- Git repository for manifests
- ArgoCD sync
- Automatic rollback on drift

Assignment 24 — Auto-Rollback on Health Failure

Configure:

- Kubernetes deployment
- Health probe
- Automatic rollback if failure

Assignment 25 — Rate Limiting at Ingress

Implement:

- Request throttling

- IP-based rate limits
- Return HTTP 429

Assignment 26 — CI Pipeline Parallelization

Optimize pipeline:

- Parallel test execution
- Cache dependencies
- Reduce runtime by 40%

Assignment 27 — Infrastructure Drift Detection

Use:

- Terraform plan automation
- Scheduled drift detection
- Slack alert on drift

Assignment 28 — Kubernetes Pod Security Standards

Implement:

- Restricted policy
- Prevent privileged containers
- Block hostPath mounts

Assignment 29 — Rolling Restart with Zero Downtime

Simulate production restart:

- Maintain 100% uptime
- Monitor availability

Assignment 30 — Automated Backup Strategy

Automate:

- RDS snapshot
- S3 versioning
- Restore validation

Assignment 31 — Container Image Signing

Implement:

- Cosign signing
- Verification before deployment

Assignment 32 — Infrastructure CI Validation

Add:

- terraform validate

- tflint
- tfsec
- Fail pipeline on issues

Assignment 33 — Kubernetes Cluster Upgrade

Upgrade:

- Minor version upgrade
- Zero downtime
- Rollback plan

Assignment 34 — Centralized Secrets Strategy

Compare:

- Kubernetes secrets
- Vault
- AWS Secrets Manager

Implement one securely.

Assignment 35 — Autoscaling Based on Custom Metrics

Scale application using:

- Requests per second

- Prometheus metrics

Assignment 36 — Multi-Region Deployment

Deploy app:

- Two regions
- Route53 failover routing
- Health checks

Assignment 37 — Blue/Green Infrastructure

Provision:

- Separate VPC per environment
- DNS cutover
- Rollback validation

Assignment 38 — Network Troubleshooting Scenario

Simulate:

- Service unreachable
- Debug using:
    - kubectl logs
    - describe

    - tcpdump
    - network policy check

Assignment 39 — Docker Image Optimization

Reduce image size by 60%.

Document optimization process.

Assignment 40 — Kubernetes Resource Optimization

Limit:

- CPU
- Memory

Prevent OOMKilled pods.

Assignment 41 — Audit Logging

Enable:

- Kubernetes audit logs
- Detect unauthorized access

Assignment 42 — Infrastructure Diagram & Documentation

Create:

- Architecture diagram
- Network flow explanation
- Security flow explanation

Assignment 43 — Chaos Engineering Test

Simulate:

- Pod deletion
- Node failure
- Traffic spike

Document system response.

Assignment 44 — CI/CD Secrets Management

Prevent secrets from:

- Being committed
- Being logged
- Being leaked

Assignment 45 — Application Observability

Implement:

- Distributed tracing

- Prometheus metrics
- Structured logs

Assignment 46 — Custom Kubernetes Operator (Basic)

Write simple operator:

- Watches CRD
- Creates pods automatically

Assignment 47 — Infrastructure Performance Benchmark

Measure:

- EC2 performance
- Network latency
- Disk IO

Compare instance types.

Assignment 48 — Kubernetes Multi-Tenancy Model

Design:

- Namespace isolation
- Resource quotas
- Network segmentation

Assignment 49 — DevSecOps Integration

Integrate:

- SAST scan
- Dependency scan
- Container scan
- Infrastructure scan

Assignment 50 — End-to-End Production-Grade Project

Build:

- Terraform infrastructure
- Kubernetes cluster
- CI/CD pipeline
- Monitoring
- Logging
- Security
- Auto scaling
- GitOps
- Disaster recovery plan

Document everything as production-ready solution.

Final Note

These 50 assignments represent:

- Real production problems
- Senior-level thinking
- Automation mindset
- Security-first architecture
- Scalability design
- High availability planning

Completing these means you are no longer a beginner DevOps engineer — you are operating with production-level maturity.